CHAPTER 3
RESEARCH METHODOLOGY

This chapter discusses the methodology used to develop **IronGrind: A Web-Based Fitness Gym Membership and Operations Management System for Iron Grind Fitness Gym in Purok Mangga, Sulongon, Dapitan City, Zamboanga del Norte**. It describes the development process, system requirements, design, data organization, and evaluation procedures. The system is intended to organize membership applications and records, membership periods and renewals, staff-recorded attendance, products, sales, and reports. Its development is organized using Agile, with work planned and delivered incrementally. Proposed evaluation activities are distinguished from results that have not yet been collected.

System Development Methodology

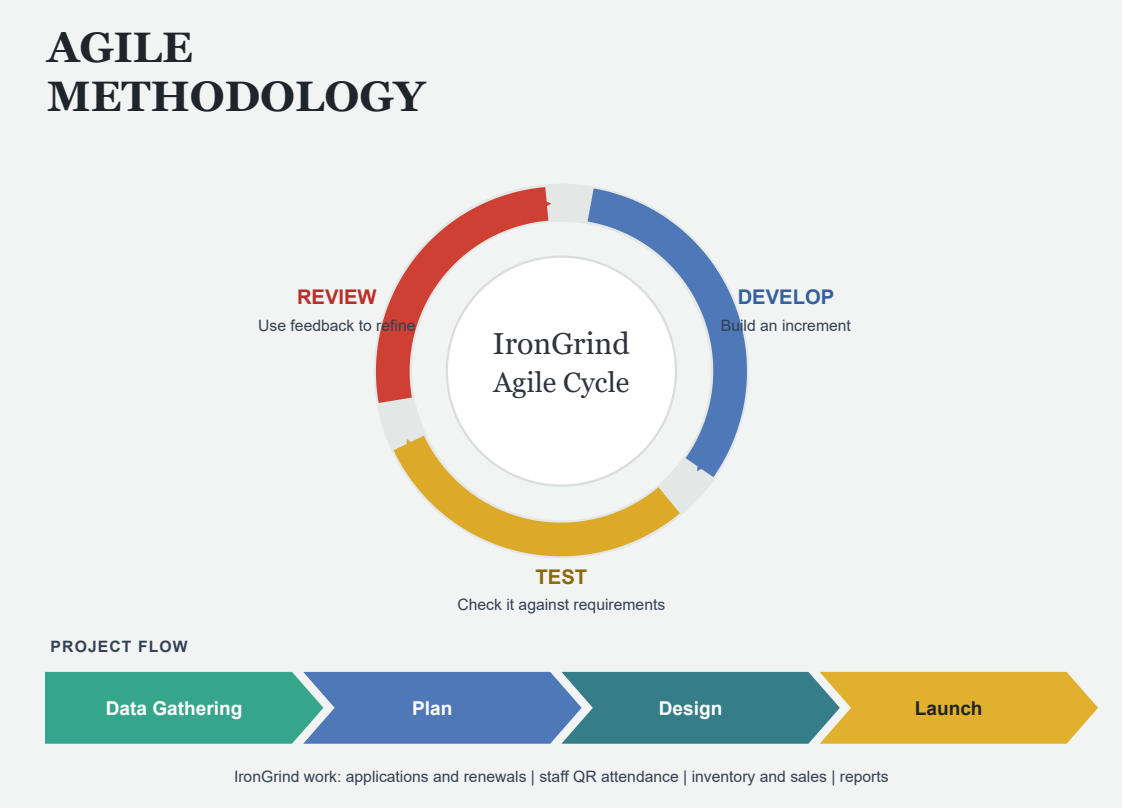


Figure 1. *Agile Methodology.*

Agile is an iterative system-development approach in which stakeholder needs are gathered and prioritized, selected work is developed and tested in increments, and review results inform the next iteration. Figure 1 has two connected parts. The circular cycle places Develop, Test, and Review around the IronGrind project name: the team builds a selected increment, checks it against its requirements, and uses review feedback to refine the next

increment. The lower project flow shows Data Gathering, Plan, and Design before Launch. This lower row represents the project's overall progression; Launch follows verification of the required increments rather than occurring at the end of every cycle. The work is organized around the gym's applicant/member and administrator workflows. The figure describes the Agile activities used to organize the project and does not prescribe a particular framework or iteration length.

Requirements Planning

Requirements planning organizes the workflows and scope described in Chapters 1 and 2 into a prioritized task list. Work begins with account access and membership applications, followed by staff review, member access, attendance, inventory, sales, and reporting. Each selected requirement is assigned acceptance criteria so the increment can be checked during construction. Privacy and authorization requirements apply throughout development because the system handles personal and health-related information.

User Design

For each iteration, the relevant user workflows, page layouts, forms, reports, data relationships, and role permissions are designed or refined. The design distinguishes pending applications from approved member records, keeps attendance as a separate operational record, and stores sales with their itemized lines. The architecture, use-case, and data-flow figures in this chapter summarize the system design.

Prototypes

The system prototype is an interactive, browser-based version of the gym workflows. Applicants use the sign-in and registration pages to enter the application workflow. Members use the member portal to review membership information, display their QR code, view attendance history, and submit an eligible renewal request. Administrators use the dashboard to review applications and manage membership, staff-scanned attendance, products, sales, receipts, and reports. The prototype uses Supabase Auth and configured database services for supported operations. Its key pages are listed in the Forms section. Deployment and operational readiness are separate implementation activities.

Construction

Selected requirements are implemented in the HTML, CSS, and JavaScript client and the Supabase authentication and PostgreSQL services. Database policies and functions support authorization and operational transactions. Each increment is checked against its acceptance criteria. Issues identified during verification are corrected or returned to the prioritized task list for a later increment.

Implementation

The implementation phase follows verification of the required increments. It includes configuring the website and Supabase project for the intended environment, applying the approved access policies, preparing administrator accounts, and orienting staff to the supported workflows. Deployment and user-orientation outcomes are reported after those activities are completed.

Research Design and Development Approach

This study produces a web-based information system for a defined operational setting. The system-development work is organized into requirements planning, user design, construction, verification, and implementation activities described above. Agile describes how the software work is organized; it is distinct from the research design used to evaluate the completed system. Chapter 1 establishes the system objectives and Chapter 2 identifies the DeLone and McLean Information Systems Success Model as a possible evaluation framework. The final formal research-design label and evaluation method should follow the design approved for the study. No experimental effect or user-evaluation outcome is claimed in this chapter.

Study Locale

The study is situated at Iron Grind Fitness Gym in Purok Mangga, Sulongon, Dapitan City, Zamboanga del Norte, as stated in Chapter 1. The system is tailored to the gym's membership and front-desk operations. Applicants and members use the web pages remotely or at a compatible device; the administrator and staff workflows are intended for the gym's authorized users. The project files do not specify a separate research-data-collection venue or schedule, so any such details should be supplied from the approved study plan.

Intended Users, Respondents, and Sampling

The system's intended users are applicants, approved members, and authorized administrators or staff. Walk-in customers are represented in staff-entered sales and do not require accounts. These user roles define who may interact with the system; they do not by themselves establish who participated in the study evaluation.

The supplied chapters and application files do not identify the final respondent population, number of participants, inclusion criteria, sampling method, or actual sample size. Those details must match the approved research proposal and the participants who actually take part. If respondents include gym members, the study should identify the eligible member group and the selection method. If the owner, administrator, or staff are included, their roles and number should be reported separately. Do not report intended users as actual respondents unless data were collected from them.

Research Instrument and Evaluation Criteria

Chapter 2 identifies the DeLone and McLean model as a possible framework. The instrument should measure only the dimensions selected and approved for this study. Its items need to match the intended respondents' experience: members can assess portal access and member-facing information, while administrators or staff can assess the operational workflows they use. Questions about features a respondent has not used should not be treated as direct experience ratings.

The approved instrument, item wording, response scale, validation process, and reliability procedure were not included in the supplied project files. The final manuscript should identify whether the study uses a questionnaire, task checklist, interview guide, or a combination; state the source or development process; describe any expert review or pilot activity that actually occurred; and append the approved instrument if required. The framework below is a mapping guide, not a substitute for the instrument.

Evaluation area	Example subject for an approved item or task	Respondent group with relevant experience
System quality	Navigation clarity, availability, error recovery, and reliable completion of common operations	Members and administrators/staff, limited to workflows they use
Information quality	Accuracy, completeness, readability, and usefulness of membership, attendance, inventory, sales, and reports	Members for their own portal data; administrators/staff for operational records
Service quality	Assistance, support, and guidance received during system orientation or use	Participants who actually received support
Use and intended use	Ability or willingness to use the functions assigned to the respondent's role	Members and administrators/staff by role
User satisfaction	Overall experience after completing representative tasks	Participants who used the system during evaluation
Net benefits	Perceived contribution to organizing and retrieving gym records	Participants with sufficient experience to judge the relevant work

Data-Gathering Procedure

The project files describe the system but do not document a completed participant evaluation. A defensible procedure should be finalized with the approved proposal before data collection. A general sequence is to obtain institutional and gym authorization, explain the study and obtain informed consent, prepare test accounts and representative tasks, orient participants without coaching their ratings, allow them to use the relevant functions, administer the approved instrument, and secure the responses for analysis. The final chapter

should state the actual dates, setting, participants, task set, instrument administration method, and handling of incomplete responses. These details should describe what was done, not a proposed procedure presented as completed work.

Statistical Treatment of Evaluation Data

The analysis method depends on the approved instrument and response scale. The proposal should specify how responses are coded, how missing answers are handled, what descriptive statistics or other analyses are used, and how rating ranges are interpreted. If a Likert-type questionnaire is used, the manuscript should state the exact scale anchors and the approved calculation and interpretation rules. The supplied project files do not provide respondent data or an approved statistical-treatment plan; therefore, this section does not report weighted means, ratings, acceptance levels, or conclusions.

Ethical and Privacy Considerations

The system handles personal identifiers, contact and emergency-contact details, and health declarations. Evaluation should use only the information required for the study, explain the purpose and voluntary nature of participation, restrict access to collected responses, and avoid exposing live member records in screenshots or demonstration accounts. Test data should be used where possible. Any consent, retention, anonymization, and disposal procedures must follow the institution's approved ethics requirements and the gym's privacy procedures. The presence of access policies in the application does not by itself establish that the research or deployment complies with every applicable requirement.

REQUIREMENT SPECIFICATIONS

This section presents the operating context and the functional, input, output, software, hardware, and nonfunctional requirements for IronGrind. These specifications describe the system represented by the current project files.

Study Setting and Intended Users

The system is intended for Iron Grind Fitness Gym in Purok Mangga, Sulongon, Dapitan City, Zamboanga del Norte. Its intended users are applicants, members, authorized administrators, and walk-in customers served by staff. The database defines `admin` and `member` profile roles; it does not define a separate staff role.

Functionality Requirements

The functional requirements describe the operations available to applicants, members, and authorized administrators. They cover sign-in, application review, membership, attendance, inventory, sales, and reports.

Table 1. Functional Requirements

ID	Requirement	Description
FR-01	Authentication and access	Users authenticate through Supabase Auth. Administrator and member access is governed by profile roles and database authorization policies.
FR-02	Online application	An applicant submits registration details through the website. The application is stored for administrator review and is not automatically an active membership. If the applicant attempts to log in while the application is pending, the website displays the pending status.
FR-03	Application review	Authorized staff view, approve, or reject registration and renewal applications. The decision and application status are retained.
FR-04	Membership management	Authorized staff manage member records, plan and membership dates, status, payment information, and membership lifecycle records.
FR-05	Member portal	An approved member views membership status and progress, personal attendance history, and a QR code; updates are limited to supported account and personal-information fields.
FR-06	Renewal request	An eligible member can submit a renewal request for staff review. A request is not an automatic renewal. Chapter 2 specifies the five-calendar-day renewal window before and including the current expiry date.
FR-07	Attendance	Staff scan a member's QR code to record check-in and check-out. Members can view their own attendance history; the member does not record their own attendance.
FR-08	Product and inventory management	Authorized staff maintain products and inventory. Supplements use gram-based quantities; merchandise uses item quantities and stock tracking.
FR-09	Sales and receipts	Staff record in-person product and membership sales for members or walk-in customers. The system stores a sale record and its line items and supports receipt/report views.
FR-10	Operational reports	Authorized staff view and print or save the available membership, member-status, archive, payment, attendance, renewal, inventory, and sales reports. Report contents depend on the records entered into the system.

Explanation of Functional Requirements

FR-01 — Authentication and access. Supabase Auth verifies account credentials and maintains the authenticated session. The application role determines which interface is presented, while database policies and server-side functions enforce access to protected records. A role displayed by the client is not sufficient on its own to authorize a database operation.

FR-02 — Online application. The application form accepts a prospective member's account and membership details. The request is persisted separately from the approved member record and remains pending for administrator review. This separation lets the system show a request's progress without treating every signup as an active membership.

FR-03 — Application review. The administrator queue presents new registration and renewal requests for decision. Approval or rejection records the status and review metadata. Registration approval creates or links a member record; renewal approval updates an existing member's current membership after retaining the prior state. A successful decision should be reflected in the member list and reports after data refresh.

FR-04 — Membership management. The administrator can view the member register and perform supported plan-change or counter-renewal actions. A current membership record contains the member's current plan and dates. Expired periods and plan changes are retained in archive/history records so that changing current values does not erase the earlier period.

FR-05 — Member portal. An authenticated member sees permitted information associated with their own account, including current membership data, QR identifier, and attendance history. Profile changes are limited to fields offered by the portal. Data access must remain scoped to the signed-in identity by database policies.

FR-06 — Renewal request. A member submits a request during the configured five-calendar-day period before and including expiry. The application remains pending until staff review. A request does not automatically change the current membership dates or count as a completed renewal transaction.

FR-07 — Attendance. Staff use the administrator scanner to resolve a QR identifier and record a visit. A scan checks for an existing open visit to determine whether the action is check-in or check-out. The member portal can display the member's own history but does not provide the staff attendance-write function.

FR-08 — Product and inventory management. The administrator maintains the catalog and can archive items. Product category determines the unit: supplements are measured by gram, while merchandise is measured by item and its available stock is tracked. Archived products are excluded from active choices while historic sale lines retain their item descriptions.

FR-09 — Sales and receipts. Staff record customer and item details for counter transactions. A sale header stores customer, receipt, date, status, and total; sale-item rows store each product or membership line. The receipt view presents those saved transaction details. For merchandise, the transaction workflow validates and updates stock.

FR-10 — Operational reports. The administrator selects a report or export period and reviews data produced by the relevant processes. Report outputs summarize saved records and their configured status/date fields. They do not establish outcomes beyond the data collected by the system.

Input Requirements

Table 2. Input Requirements

Input requirements identify the information supplied to the system by applicants, members, and authorized administrators or recorded through staff actions. Required fields depend on the selected workflow.

Input requirement	Description
Account credentials	Email and password are submitted to Supabase Auth.
Registration or renewal application	Applicant and plan details, contact and emergency-contact information, goals, health declarations, and signature where collected by the form.
Attendance action	Staff scans a member QR code and records check-in or check-out.
Product and inventory update	Administrator enters product details, price, unit, and merchandise stock quantity where applicable.
Sale transaction	Staff enters customer type and name, products or membership, quantities, and transaction details.
Report request	Administrator selects a report type and available filters or date range.

Input Design

Input design specifies how information is collected and checked before it becomes an application, member update, attendance record, product, or sale. The browser forms provide immediate validation to help a user correct incomplete or incorrectly formatted entries. Database constraints and authorized functions provide additional checks for shared records. The following table maps the main input groups to their validation and destination.

Input group	Examples of captured values	Validation or processing	Main destination
Sign-in	Email/username and password	Required credential fields; Auth validates account and session	Supabase Auth; `profiles` role lookup
Registration request	Name, contact, plan, membership dates, declarations, signature	Required fields, supported formats, plan selection, unique account identifiers	`membership_applications`
Renewal request	Existing member, selected plan, requested details	Existing member link and configured eligibility window; staff review remains required	`membership_applications`

Input group	Examples of captured values	Validation or processing	Main destination
Staff decision	Application identifier and approve/reject action	Administrator authorization; pending status checked before decision	`membership_applications`, `members`, `membership_history`, and `sales` as applicable
Attendance scan	QR payload and staff scan action	Resolve identifier; check open visit and expiry before check-in	`attendance`
Product maintenance	Name, category, unit price, unit, size, stock, optional image	Name required; positive valid price; nonnegative stock; category-unit consistency	`products`
Sale entry	Customer type/name, product or membership line, quantity	Active product, positive quantity, merchandise integer/stock limit, valid membership line	`sales` and `sale_items`; product stock where applicable
Report request	Report type and optional month/date	Supported report identifier and valid year-month input for monthly export	Report query and generated output

Inputs containing emergency contact or health information are personal data. The system should collect only the values required by the approved workflow, restrict access through the deployed policies, and follow the gym's established privacy and retention procedures.

Output Requirements

Table 3. Output Requirements

Outputs are the responses and records produced after the system validates a request and processes the stored data. Email notifications remain planned and are not currently sent.

Output requirement	Description
Authentication and login result	Role-appropriate routing or an authentication error; pending applicants see their application status during login.
Application and membership result	The recorded application status and, after staff review, the decision and updated membership information.
Member portal information	Membership details, QR code, personal attendance history, and active product catalog permitted for the member.
Attendance confirmation	Confirmation of staff-recorded check-in/check-out and permitted attendance history.

Output requirement	Description
Receipt	Sale details, line items, quantities, prices, customer information, and total.
Administrative reports	Membership, member status, archives, payments, attendance, renewals, inventory, and sales data based on stored records.
Planned: email notifications	Application or renewal decisions and expiry reminders are planned; email delivery is not currently available.

Output Design

Output design presents the result of a request in a form appropriate to its user. Applicants receive a submission or status response; members receive only their permitted membership and attendance information; administrators receive management views, transaction receipts, and reports. The table identifies the content and intended use of principal outputs.

Output	Recipient	Information presented	Purpose
Login result	Applicant/member/administrator	Successful role routing, pending application status, or authentication error	Tell the user whether the requested account workflow can continue
Application status	Applicant/member and administrator	Pending, approved, or rejected status with review information where available	Communicate the application decision and queue state
Member portal view	Signed-in member	Current plan, dates, time-based progress, QR identifier, and own attendance history	Let the member review their recorded information
Attendance confirmation	Staff	Member name and check-in/check-out result or error	Confirm a scan operation and prompt staff to retry when needed
Transaction receipt	Customer and administrator	Receipt number, customer, date, item descriptions, quantities, unit prices, line totals, and transaction total	Provide a transaction record for a completed sale
Operational report	Administrator	Data selected for membership, status, archives, payment, attendance, renewals, inventory, or sales	Support review of stored operational records
Monthly report archive	Administrator	PDF reports for the selected month in a ZIP download	Provide portable report copies for the chosen period

Output	Recipient	Information presented	Purpose
Email notification	Planned recipient	Approval or expiry notice	Future communication; no configured provider currently sends it

Software Requirements

Table 4. Software Requirements

Software	Requirement or purpose
Web browser	Current browser with JavaScript enabled for applicant, member, and administrator workflows.
Frontend	HTML, CSS, and JavaScript files served by a compatible web host.
Backend services	Supabase Auth and PostgreSQL, with the project schema, Row Level Security policies, and required database functions configured.
Edge Function	Supabase Edge Function runtime for administrator-authorized member account creation.
Development environment	Node.js 20 or later to run the repository's local static-file server. This local server is for development, not the production database backend.
QR scanning	Browser camera permission and a secure HTTPS origin for production QR scanning.

Hardware Requirements

Table 5. Hardware Requirements

User or component	Requirement
Administrator or staff	Desktop computer, laptop, or compatible mobile device with a current browser and internet access. A camera-enabled device is needed when staff scan member QR codes.
Applicant or member	Smartphone, tablet, laptop, or desktop computer with a current browser and internet access.
Web hosting	Hosting capable of serving the static HTML, CSS, JavaScript, and image assets over HTTPS. Supabase provides the separate authentication and database services.

Nonfunctional Requirements

Table 6. Nonfunctional Requirements

Area	Requirement
Platform	The application is accessed through a modern browser with JavaScript enabled. The user interface is implemented as web pages and client-side scripts.
Connectivity	Authenticated database-backed functions require an internet connection to reach Supabase.

Area	Requirement
QR scanning	The staff device requires a camera and browser permission to scan QR codes. Production camera access should be served from a secure HTTPS origin.
Access control	Database access is restricted through authenticated roles and PostgreSQL Row Level Security policies. Privileged account creation is routed through the administrator-verified Edge Function. The deployed configuration must be tested before use.
Data handling	The application stores contact, emergency-contact, and health-related information. Access and retention practices should follow the gym's approved privacy procedures and applicable law. This description is not a certification of legal compliance.
Availability	Users depend on the availability of the website host, internet connection, and Supabase services. Operational backup and recovery procedures must be defined for the deployed environment.

SYSTEM DESIGN

System Architecture Diagram

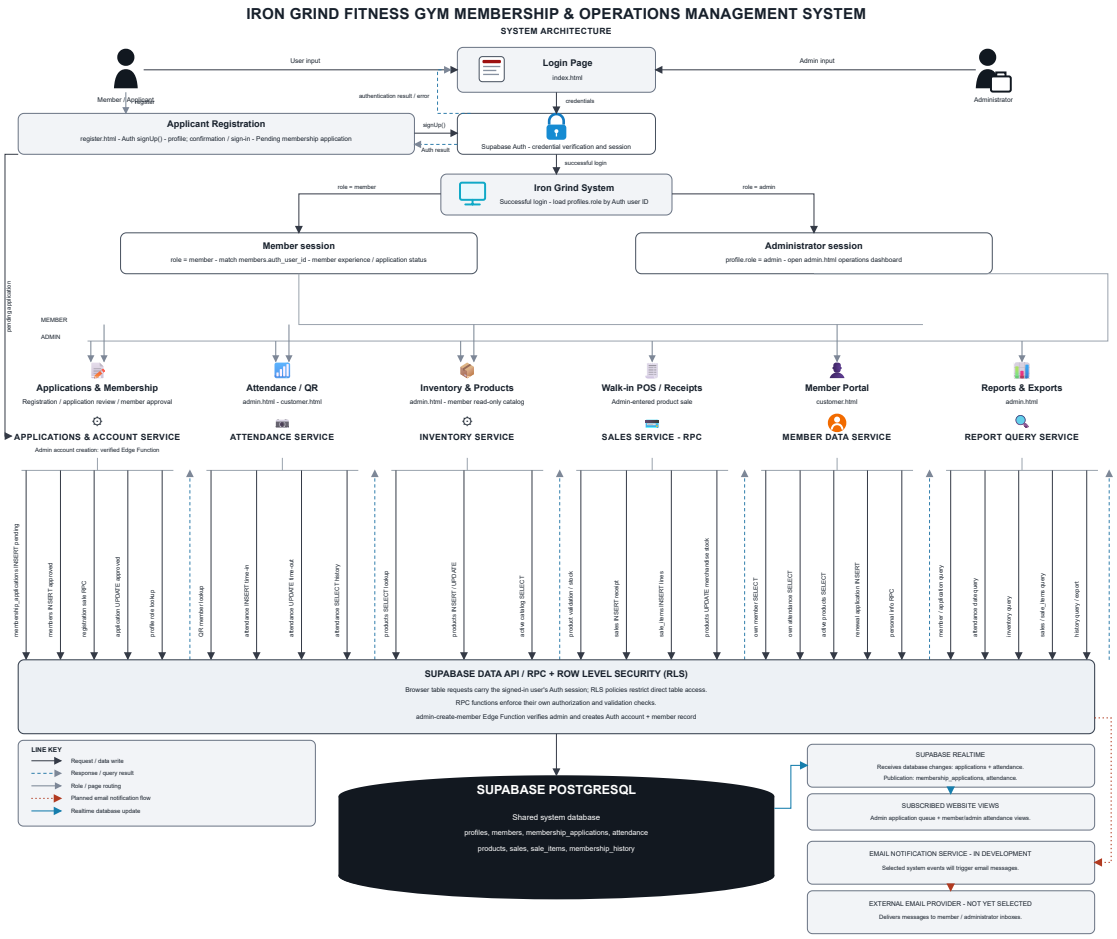


Figure 3. System Architecture Diagram

Figure 2. *System architecture of IronGrind.*

Figure 2 is a component view of the system and shows where each kind of work takes place. On the user side, applicants and members use the login, registration, and member pages; administrators use the dashboard. Supabase Auth verifies credentials and establishes a session, while the application uses the profile role to route a signed-in person to the appropriate interface. Inside the browser application, the labeled modules group related functions: Applications and Membership handle submissions and staff decisions; Attendance and QR supports staff scans; Inventory and Products maintains the catalog; Walk-in POS and Receipts records transactions; Member Portal presents a member's permitted information; and Reports and Exports prepares administrative summaries. Requests go to Supabase through its Data API or database RPC functions. The PostgreSQL database contains the application records, and Row Level Security (RLS) policies plus function-level checks restrict access to them. Realtime sends changes from `membership_applications` and `attendance` to subscribed website views. The email component is marked as in development because no external provider is selected. The Node.js server in the repository serves static files for local development and is not the production database backend.

Explanation of Architecture Components

Component	Responsibility	Main interaction
Applicant interface	Presents sign-in and membership registration pages.	Sends credentials to Auth and application details to the database-backed application workflow.
Member portal	Presents permitted membership, attendance, QR, catalog, and account information.	Reads member-scoped records and submits supported profile or renewal requests.
Administrator dashboard	Presents operational summaries and administration sections.	Reads and updates records through authenticated database requests and configured functions.
Supabase Auth	Verifies credentials and issues user sessions.	Supplies the identity used by profile lookup and database policies.
Supabase Data API	Provides authenticated access to PostgreSQL tables and views.	Applies database role and RLS rules to client requests.
Database RPC functions	Perform selected privileged or transactional workflows.	Validate caller authorization and carry out operations such as recording a sale or membership change.
PostgreSQL public schema	Stores profiles and gym operational records.	Enforces keys, relationships, checks, and unique constraints defined in the deployed schema.
Realtime service	Announces configured database changes to subscribed views.	Current admin subscriptions refresh application and attendance information.
Static web host	Serves HTML, CSS, JavaScript, and images.	Delivers the browser application over HTTPS in the deployed environment.

Component	Responsibility	Main interaction
Email integration	Planned external communication component.	Not active because no provider is configured in the current project.

The architecture separates interface responsibilities from record storage and authorization. The browser gathers user input and renders results, but shared data is stored in PostgreSQL. Authentication establishes who is making a request; RLS and checked database functions determine what that identity can do. Static hosting and the database service are separate deployment components, so a browser page may load successfully while a database request still fails due to connectivity, session, policy, or schema configuration.

Data Process Modeling

Data process modeling describes how information enters IronGrind, which processes handle it, where records are stored, and what results are returned to applicants, members, and administrators. The model is based on the current Supabase schema and implemented workflows. Email notification is shown only as a planned feature.

Diagrams

The diagrams that follow present the system at different levels. The context diagram defines the system boundary and external exchanges. The Level 1 data-flow diagram expands the system into processes and stores. The use-case diagram groups functions by actor, and the ERD describes the database relationships that support those functions.

Context Diagram

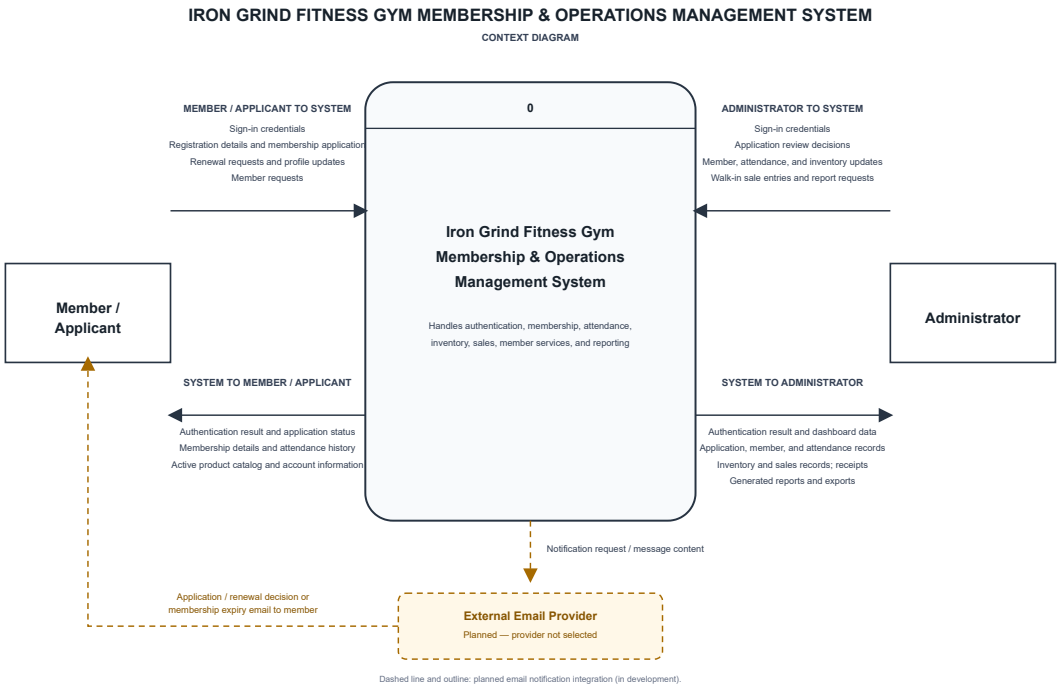


Figure 3. *System context diagram of IronGrind.*

Figure 3 sets the boundary around IronGrind as a single system, without showing its internal modules or database tables. Applicants and members send credentials, registration or renewal details, and supported profile requests into that boundary. IronGrind returns authentication results and the membership, attendance, or active catalog information their role permits. Administrators send review decisions, member and product updates, sale entries, and report filters; the system returns updated records, receipts, and reports. A walk-in customer is represented through the staff-entered purchase rather than a direct system account. Supabase Auth and Supabase data services appear outside the application boundary because they provide authentication and database services to the website. The email exchange is planned and should be read as a future integration, not a current system output.

Data Flow Diagram

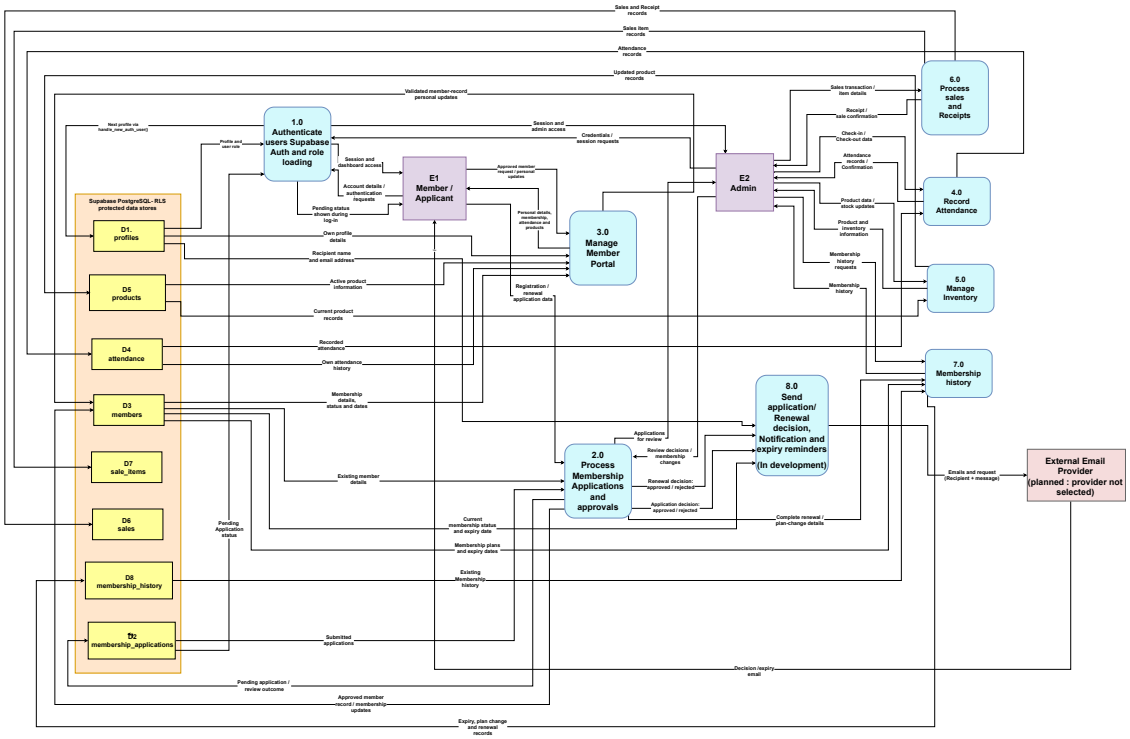


Figure 4. *Level 1 data-process model for IronGrind.*

Figure 4 decomposes the single IronGrind system from Figure 3 into numbered processes and named data stores. External actors provide inputs to the process that handles them: credentials go to authentication; application details and staff decisions go to application processing; member requests go to the portal; QR scans go to attendance; product changes go to inventory; and customer and item details go to walk-in sales. The report process reads the operational stores to prepare summaries. The data stores D1–D8 correspond to profiles, membership applications, members, attendance, products, sales, sale items, and membership history, as listed in the Data Stores table. Arrows between processes and stores distinguish information read from information written; arrows between a process and an actor represent the corresponding request or response. The figure's email process is explicitly planned only: the current system has no configured email provider and does not send these messages. Together, this diagram explains the record flow in more detail than the context diagram, while the tables below define each process and store in text.

Explanation of Data-Flow Paths

Identity path. Credentials move from the user to the authentication process and Supabase Auth. After authentication, the profile role is read to determine the appropriate application route. Database requests then carry the authenticated identity so policies can evaluate the request.

Application path. Applicant details enter the application process and are stored in D2. Administrator decisions read the pending application and write its status and review metadata. An approved registration produces or links a D3 member record; a renewal uses the existing D3 member. Related transactions and lifecycle events are recorded in D6/D7 and D8 when the corresponding operation succeeds.

Member portal path. The signed-in member's request is checked against their identity and permitted data access. The portal reads D3 for current membership information, D4 for attendance history, D5 for the active catalog, and D1 for profile details where needed. A renewal request writes to D2 and awaits an administrator response.

Attendance path. A staff scan supplies a QR identifier to process 4.0. The process reads D3 to resolve the member and D4 to determine whether an open attendance visit exists. It writes a new check-in or updates the open row with a check-out timestamp, then returns confirmation to staff.

Inventory and sales path. An administrator's product changes write to D5. A counter sale reads D5 to validate the selected product and available merchandise quantity, then writes a transaction header to D6 and its line to D7. The sale result is used to display a receipt and refresh operational views.

Reporting path. The report process reads the data stores relevant to the selected report, applies the configured date/status grouping, and returns a view or export to the administrator. It does not change source operational records as part of ordinary report viewing. Email is not a live path in the current deployment.

Data Process Explanations

The Level 1 diagram can be read as a sequence of related responsibilities. Each process validates its input, uses only the records needed for its task, and returns the corresponding result.

1.0 Authenticate User. Supabase Auth verifies submitted credentials and provides the authenticated session. The application reads the associated profile role to route the account to the member or administrator interface. Pending applicants receive an application-status response; they are not treated as approved members.

2.0 Process Membership Applications. The system records registration or renewal details in the applications store. An authorized administrator reviews the request. Approval creates or updates the member record and records the associated transaction where configured; rejection updates the application status.

3.0 Manage Member Portal. A signed-in member requests permitted membership, attendance, catalog, or profile information. The system returns records allowed for that member and stores a renewal request for staff review.

4.0 Record Attendance. Staff scan a member QR code. The system resolves the member and either records a check-in or closes the member's open visit. The QR code identifies a member for staff use; it does not grant entry by itself.

5.0 Manage Inventory. An authorized administrator creates, edits, or archives product records. Supplement prices and quantities use gram units; merchandise uses item units and tracked stock. The active catalog is returned to permitted views.

6.0 Process Walk-in Sale. Staff enter the customer and selected product. The system validates product availability, records a sale header and itemized line, updates merchandise stock according to the configured transaction, and provides receipt details.

7.0 Query Reports and History. An administrator selects a report and available filters. The system reads the relevant membership, attendance, inventory, sales, or history records and returns the corresponding view or export.

8.0 Email Notifications (planned). Approval and expiry events may be used by a future notification integration. No external provider is configured, so this process currently produces no email output.

Operational Workflow Details

The numbered processes above summarize the flow. The following workflow descriptions state the initiating actor, the information checked, the records affected, and the observable result. They are intended to guide implementation review and functional verification.

Account Access Workflow

An applicant, member, or administrator submits credentials through the sign-in page. Supabase Auth validates the credentials and returns an authenticated session. The application then retrieves the profile associated with that Auth identity and uses its stored role for navigation and role-specific actions. An applicant with a pending registration is not treated as a member merely because an Auth account exists. If the credentials are invalid, the Auth service is unavailable, or the profile cannot be retrieved, the system returns an error or status message and does not claim that access succeeded. Database access remains subject to RLS even after the client routes to a page.

New Membership Application Workflow

An applicant creates an account and completes the application form. The client checks required fields and basic formatting, including the supported contact and password formats. The submission includes requested plan information and, where collected, demographic, emergency-contact, health-declaration, goal, and signature values. The application is associated with the Auth identity and begins in pending status. The member record is not considered active while this review is pending. An authorized administrator opens the registration queue and checks the submitted details. Approval creates or links the member record and records the applicable membership transaction; rejection updates the application review state. Review time and reviewer identity are stored when available. A repeat submission or interrupted approval should be checked against existing records to avoid creating duplicate member links or receipt records.

Renewal Request and Approval Workflow

An existing member opens the renewal request control when eligible under the configured renewal window. The client submits the selected plan and membership details as an application associated with the existing member. That request remains pending while the current membership period remains the authoritative member record. An administrator reviews the renewal queue and approves or rejects it. On approval, the system records prior and new plan dates and prices in membership history, updates the member's current plan and expiry, and records the renewal sale and item line. The transaction should be written in a retry-safe order so an interrupted request can be resumed without charging twice or leaving inconsistent membership dates. The report distinguishes a recorded receipt from a legacy approved request for which a receipt row is missing.

Staff Attendance Workflow

Staff open the attendance section and grant camera permission to the browser. The scanner decodes the member identifier and resolves it against the loaded member records or Supabase. The process first checks for an open visit for that member. If one exists, the scan records checkout; otherwise, the membership expiry is checked before a new check-in is created. The stored row includes the member reference, timestamps, and the staff identity where supported. Repeated checkout requests must not overwrite a completed visit. If QR resolution, network access, or the database write fails, staff receive a clear failure message and can retry. The attendance list and dashboard are refreshed from the resulting record.

Inventory Maintenance Workflow

An administrator creates or edits a product by entering its name, category, unit price, and applicable variant or stock data. Validation rejects an empty name, invalid price, or invalid stock value. Category determines the selling unit: supplements are per gram, while

merchandise is per item and has countable stock. An archive action changes the product's active state and records when it was archived; archived products are removed from new sale choices but remain visible in the archive report. A product image is optional and is resized by the client before it is persisted. The browser interface is not the authorization boundary; database policies must also limit product mutations to administrators.

Counter Sale Workflow

Staff enter the customer name, choose an active product, and specify its quantity. Merchandise quantities are whole items and must not exceed the available stock. Supplement quantities are gram amounts and can use decimal precision. The database transaction records a receipt header and its sale line; configured sale functions validate current stock and update merchandise quantities. Once the saved result is returned, the interface updates the sale list, inventory summary, and receipt preview. If a write fails, a receipt should not be shown as completed. Receipt number uniqueness and transaction-level validation help prevent duplicate or inconsistent sale records.

Membership Plan Change Workflow

An administrator selects a current member and a replacement plan, reviews the new charge, and submits the change. The system validates the member link, plan, positive price, and effective dates. It stores a plan-change history event with the prior plan and dates and the new plan and dates, then updates the current member row. Where the workflow records a charge, it also creates a membership sale and receipt line. The displayed confirmation follows successful database completion. In the local demonstration flow, corresponding browser storage records are updated; the Supabase flow uses the configured database function and should be used for shared production records.

Report Generation Workflow

An administrator selects a report type and, for monthly exports, a valid year and month. The report process reads the applicable data stores and formats the selected fields. Date-based reports group or filter records using the stored event date and display times in the local gym context. Inventory and member-status output represent current values unless a historical snapshot is available. PDF and ZIP generation depends on the required browser libraries loading successfully. The interface reports an unavailable generator or invalid month rather than claiming a file was created.

Use-Case Diagram

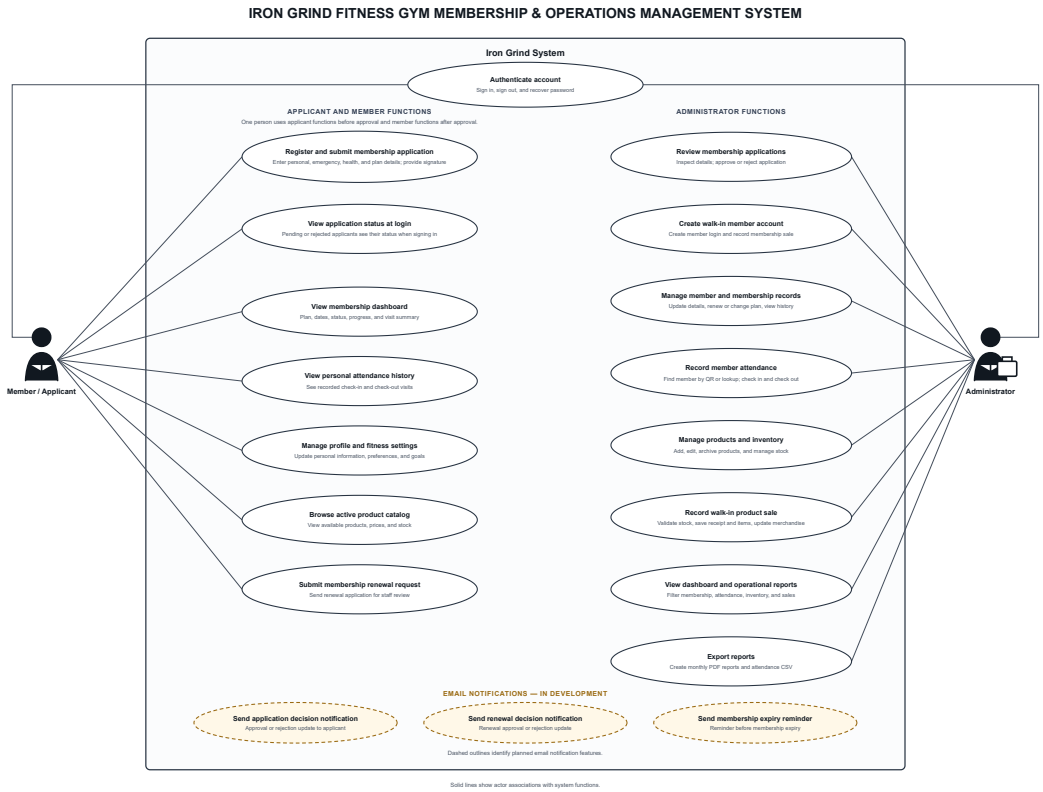


Figure 5. *Use-case diagram of IronGrind.*

Figure 5 describes the system from each actor's point of view. The actor lines connect a role to the use cases that role may initiate; the system boundary encloses the functions provided by IronGrind. Applicants can register and access the application/login workflow. Members can use the member portal, view their membership and attendance information, and submit renewal requests when eligible. Administrators review applications and manage members, attendance, products, sales, and reports. Walk-in customers are not shown as authenticated users because staff enter their sale details at the counter. The diagram expresses available role interactions; it does not imply that a member can approve an application or record their own attendance.

Use-Case Explanations

The use cases in Figure 5 are grouped by the actor's role. The associations show which actor can initiate each function; database policies and server-side checks still control access to the underlying records.

Applicant. An applicant creates an account and submits a membership application. The application remains pending until an administrator reviews it.

Member. A member signs in to view permitted membership details, the issued QR code, personal attendance history, and the active catalog. An eligible member may submit a renewal request, which remains subject to staff review.

Administrator. An administrator reviews applications and manages member records, staff-scanned attendance, products, sales, receipts, and operational reports.

Walk-in customer. A walk-in customer does not need a system account. Staff enter the customer's purchase in the point-of-sale workflow and provide a receipt.

Detailed Use-Case Descriptions

Each use case describes a user goal and its normal result. The alternate paths identify conditions the interface must handle; they are not claims that a particular test has already been executed.

Use Case 1 — Submit Membership Registration

Element	Description
Primary actor	Applicant
Preconditions	The registration page is available; the applicant can provide the required account and application information.
Main flow	The applicant creates or uses an Auth account, completes required fields, selects a plan, submits the application, and receives a pending status.
Alternate flow	Invalid or missing fields are shown for correction; duplicate account details or a service error prevent successful submission and produce an error message.
Postcondition	A pending registration application is associated with the submitting identity; no active member record is implied before approval.

Use Case 2 — Review and Decide an Application

Element	Description
Primary actor	Administrator
Preconditions	The administrator is authenticated and authorized; the request is pending.
Main flow	The administrator opens the appropriate queue, reviews submitted details, and approves or rejects the request. The system records the decision and review metadata.
Alternate flow	If the request is no longer pending or the database operation fails, the system reports the condition and refreshes the queue before retry.
Postcondition	A rejected request remains recorded as rejected; an approved registration is linked to a member record and associated sale according to the configured workflow.

Use Case 3 — Submit and Approve a Renewal

Element	Description
Primary actor	Member submits; administrator decides
Preconditions	The member has an existing member record and is within the configured renewal eligibility period.
Main flow	The member submits a renewal request. The administrator reviews it and approves or rejects it. On approval, prior and new membership values, the current membership update, and the transaction are recorded.
Alternate flow	If the member is ineligible, submission is unavailable or refused. If a request is already decided, the pending-only update should not apply again.
Postcondition	The request has a recorded decision; only an approved renewal updates current membership dates.

Use Case 4 — Record Member Attendance

Element	Description
Primary actor	Administrator/staff
Preconditions	Staff is signed in with required permissions; the device camera is available and permission is granted.
Main flow	Staff scans a member QR. The system resolves the member, checks for an open visit, then records check-in or check-out and displays the result.
Alternate flow	An unknown code, expired membership at check-in, camera failure, or database/network error is reported. An existing open visit may still be checked out.
Postcondition	A timestamped attendance event is stored or the user is told that no write occurred.

Use Case 5 — Maintain Products and Record a Sale

Element	Description
Primary actor	Administrator/staff
Preconditions	Staff is authenticated and authorized; the selected product is active.
Main flow	Staff create or update catalog data as needed, enter a customer and quantity, complete the transaction, and present the generated receipt.
Alternate flow	Invalid price/stock, insufficient merchandise quantity, inactive product, or failed transaction prevents completion and is reported.
Postcondition	The sale and item line are recorded; merchandise stock is changed by the configured transaction; a receipt can be retrieved by its receipt number.

Use Case 6 — Review Operational Reports

Element	Description
Primary actor	Administrator
Preconditions	Administrator access is active and relevant source data is available.
Main flow	The administrator selects a report, reviews the rows and totals, and optionally prints or exports the output.
Alternate flow	Empty result sets display an empty-state message; invalid month selection or unavailable PDF/ZIP libraries is reported.
Postcondition	The report is displayed or an export file is generated from available stored records.

Entity Relationship Diagram

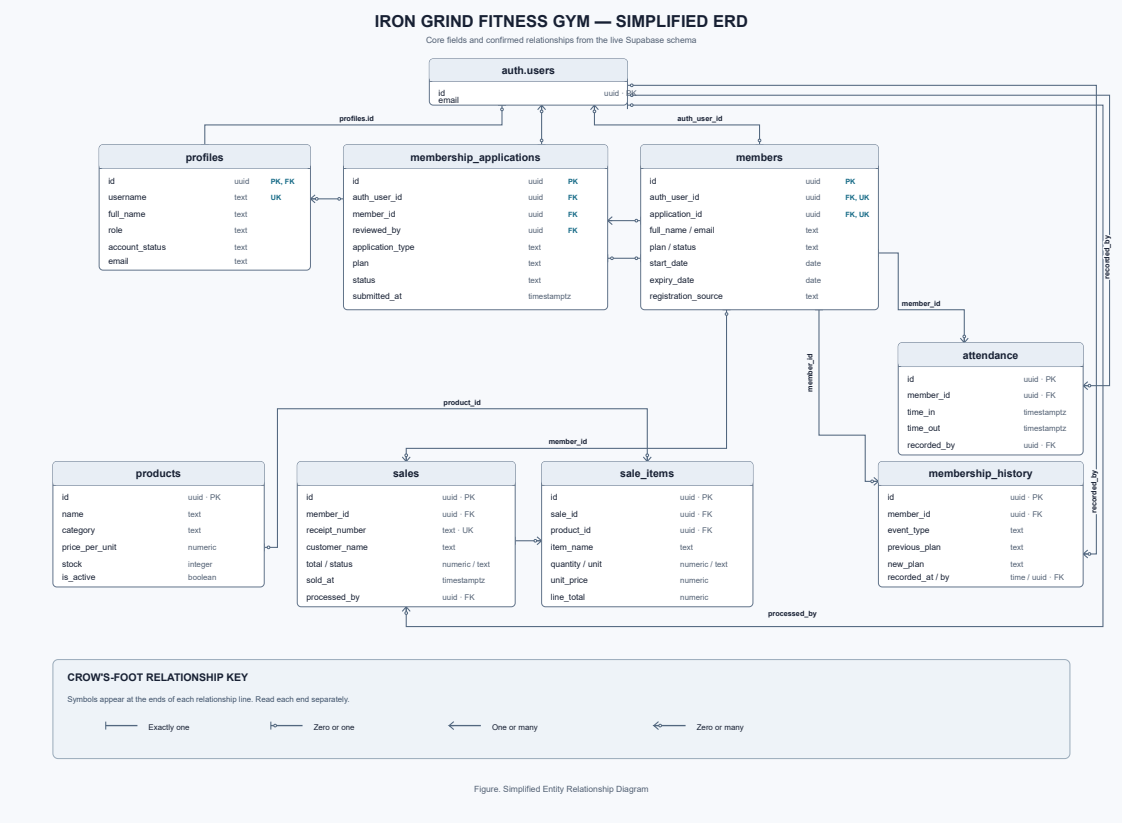


Figure 6. *Entity relationship diagram of IronGrind.*

Figure 6 is a relational view of the core tables, rather than a flow of screens or activities. Each box names a table and shows selected fields; PK marks a primary key, FK marks a foreign key, and UK marks a unique field or link. The line ends indicate relationship cardinality: for example, one member may have many attendance events and membership-history records, while each such record points back to one member. A sale may belong to a member or to a walk-in customer, and its itemized lines are stored separately in `sale_items`; a line can retain its item description even when it has no linked product. `auth.users` is

managed by Supabase and is shown to explain identity links, while `profiles` stores application role information. The `members.auth_user_id` link is nullable and unique, so a member can exist without a linked Auth identity and a single Auth user cannot be linked to multiple member rows. Application and member links also allow pending submissions to exist before they are connected to an approved member. The simplified ERD omits secondary fields to keep the relationships readable; consult the data dictionary and current schema for the full field list.

Entity and Relationship Explanations

The ERD is a database structure diagram. Primary keys identify rows; foreign keys connect records; unique keys prevent duplicate identity or receipt links. Optional links are nullable, so a record can exist before an associated account or member has been linked.

Authentication and profiles. Supabase manages authentication identities in `auth.users`. The `profiles` row stores the application role and account information. A profile's identifier corresponds to its Auth user.

Applications and members. `membership_applications` stores registration and renewal requests and their review state. An application may refer to a member after the account is approved or when it represents an existing member's renewal. `members` stores approved membership information. The nullable unique `members.auth_user_id` link permits no more than one member row per Auth user.

Attendance and membership history. Each attendance row refers to the member whose visit it records; a member can have multiple visits. Membership history stores lifecycle events such as expiration, renewal, and plan change, with prior and new values where available.

Sales, sale items, and products. A sale stores the receipt-level customer, date, status, and total. Its itemized details are stored in `sale_items`. A line may refer to a product, while the item name and price are retained on the line for the transaction record. A member sale may link to a member; a walk-in sale may have no member link.

Profile-to-identity relationship. Each application profile uses the same UUID as its Auth identity. The Auth service owns credentials, while the `profiles` table owns the application role and display information. Deleting an Auth identity cascades to its profile; application records reference Auth identities according to their defined deletion rules.

Application-to-member relationship. Applications are stored independently from members so a pending request exists before approval. A registration approval can link the resulting member to its application. A renewal application links to an existing member. The

Auth link on a member is nullable and unique, allowing an imported or staff-created member row without a portal identity while preventing two member rows from sharing one Auth user.

Member-to-attendance relationship. One member may have many attendance rows, and each attendance row belongs to one member. The foreign key is required and restricts deletion while attendance history remains. Each row stores a time-in and optional time-out; the check constraint requires the checkout not to precede check-in.

Member-to-membership-history relationship. One member may have many lifecycle events. Each event records its type and any available prior/new plan, price, and dates. This one-to-many link preserves the sequence of expiration, renewal, and plan-change records for a member.

Sale-to-sale-item relationship. One sale header can have multiple item lines. Each line must point to a sale, and deletion of that sale cascades to its lines. Membership sales use a membership line category and do not need a catalog product reference.

Product-to-sale-item relationship. One product may be referenced by many sale lines. The line's `product_id` is nullable and is set to null if a product row is removed; the line retains its item name, unit, and price so the historical receipt remains interpretable. Product archival marks the catalog row inactive rather than removing it from active selection.

Administrator-to-operational-record relationships. Review, attendance, sales, and history rows can retain the Auth user who recorded or processed the event. These audit references may become null if the associated Auth identity is removed, while the operational event itself remains available for historical reporting.

These links support the reports and workflows described in the process model. The simplified diagram omits secondary fields for readability; the Data Dictionary lists the principal fields and their purposes.

Data Processes

Process	Input	Data stores or services used	Output
1.0 Authenticate User	Member/applicant or administrator credentials	Supabase Auth; D1 `profiles` for role lookup	Authenticated session and role-based routing, or an authentication error.
2.0 Process Membership Applications	Registration or renewal details; administrator approval or rejection	D2 `membership_applications`; D3 `members`; D8 `membership_history`; D6 `sales` when a membership transaction is recorded	Updated application status, member record and related transaction result.

Process	Input	Data stores or services used	Output
3.0 Manage Member Portal	Member requests, supported profile updates, renewal submissions	D1 `profiles`; D2 `membership_applications`; D3 `members`; D4 `attendance`; D5 `products`	Member's permitted membership, attendance, and active product information; renewal or update result.
4.0 Record Attendance	Staff QR scan and check-in/check-out action	D3 `members` for member lookup; D4 `attendance` for attendance records	Saved attendance event and confirmation; attendance history for permitted member access.
5.0 Manage Inventory	Administrator product, price, category, or stock changes	D5 `products`	Updated product and inventory records; active catalog data for members.
6.0 Process Walk-in Sale	Staff-entered customer and sale-item details	D5 `products` for validation and stock; D6 `sales`; D7 `sale_items`	Validated sale, receipt details, and applicable merchandise stock update.
7.0 Query Reports and History	Administrator report type and filters, including date range	D2 `membership_applications`; D3 `members`; D4 `attendance`; D5 `products`; D6 `sales`; D7 `sale_items`; D8 `membership_history`	Dashboard summaries, filtered report data, and available exports.
8.0 Email Notifications (planned)	Future approval or expiry events	No live notification store or external provider configured	No email output is currently produced; this process remains planned.

Data Stores

ID	Data store	Contents
D1	`profiles`	Authenticated user's application profile and role.
D2	`membership_applications`	Registration and renewal submissions, review status, and decision information.
D3	`members`	Approved member and membership records.
D4	`attendance`	Staff-recorded member check-in and check-out events.
D5	`products`	Product catalog, categories, prices, units, and merchandise stock.
D6	`sales`	Transaction headers and receipt details, including membership sales where recorded.

ID	Data store	Contents
D7	`sale_items`	Itemized transaction lines.
D8	`membership_history`	Expiration, renewal, and plan-change events.

OVERALL DESCRIPTION

IronGrind is a browser-based membership and operations system for applicants, members, and administrators. Users submit requests through the website; Supabase Auth manages authentication and sessions; authorized data requests pass through the Supabase Data API or database RPC functions; and PostgreSQL stores operational records. RLS and function-level checks restrict access. The system supports application review, member records, staff-recorded attendance, inventory, sales and receipts, and administrative reports. Realtime subscriptions currently cover membership applications and attendance. Email notifications remain planned and are not operational.

DATA DICTIONARY

This data dictionary summarizes the public-schema fields and relationships defined in the project SQL. Supabase Auth manages credentials separately from these application tables.

Entity	Key fields and data types	Purpose and relationships
`profiles`	`id` UUID (PK, Auth FK); `email`, `username`, `full_name`, `role`, `account_status` TEXT; timestamps	User profile and role used by the application. `username` is unique.
`members`	`id` UUID (PK); `auth_user_id` UUID (nullable, unique Auth FK); `application_id` UUID (nullable, unique FK); `full_name`, `email`, `username`, `mobile` TEXT; `date_of_birth` DATE; `age` SMALLINT; `height`, `weight_kg` NUMERIC; `height_unit`, `sex`, `plan`, `status`, `registration_source` TEXT; `emergency_contact` JSONB; `plan_price`, `paid` NUMERIC; membership dates; `fitness_goals`, `health_conditions` TEXT[]; `other_medical_conditions`, `signature_path` TEXT; timestamps	Approved member and membership record. Referenced by applications, attendance, sales, sale items, and membership history.

Entity	Key fields and data types	Purpose and relationships
membership_applications	id UUID (PK); auth_user_id UUID (required Auth FK); member_id UUID (nullable FK); application_type, status, applicant and plan details; demographic, health, and signature fields; submission/review timestamps; reviewed_by UUID (Auth FK represented through the profile identity)	Registration and renewal requests and their review state. member_id can be empty before a member is linked.
attendance	id UUID (PK); member_id UUID (FK); time_in, time_out, created_at TIMESTAMPTZ; recorded_by UUID (Auth FK)	Staff-recorded check-in and check-out. A member may have multiple attendance records.
products	id UUID (PK); name, category, unit, size, image_path TEXT; price_per_unit NUMERIC; stock INTEGER; is_active BOOLEAN; timestamps	Product catalog and inventory. Supplements use gram units; merchandise uses item units and stock.
sales	id UUID (PK); receipt_number TEXT (unique); customer_type, customer_name, status TEXT; member_id UUID (nullable FK); total NUMERIC; sold_at TIMESTAMPTZ; processed_by UUID (Auth FK)	Sale header and receipt information. A sale may be associated with a member or a walk-in customer.
sale_items	id UUID (PK); sale_id UUID (FK); product_id UUID (nullable FK); item_name, category, unit TEXT; quantity NUMERIC; unit_price, line_total NUMERIC	Itemized sale lines. Product reference can be null; item name and price are stored on the sale line.
membership_history	id UUID (PK); member_id UUID (FK); event_type, previous/new plan, notes TEXT; previous/new prices NUMERIC; previous/new dates DATE; recorded_at TIMESTAMPTZ; recorded_by UUID (Auth FK)	Expiration, renewal, and plan-change history for a member.

Detailed Field Reference

The following field reference expands the summary above. It describes the principal fields in the project schema and their role in the system. PK identifies a row, FK connects it to another table or Auth identity, and UK denotes a uniqueness rule. A nullable field may be absent when

its value is not yet known or does not apply. Database defaults and checks are noted where they affect stored values. Supabase Auth credentials are managed in the separate `auth.users` schema and are not repeated here.

`profiles` — Account Profile and Authorization

The profile row extends the Auth identity with information used by the application. Its `id` shares the Auth user's UUID, so the profile is created and removed in step with the corresponding identity. The role is constrained to the application roles `admin` and `member`; the role is not taken from arbitrary signup input. Profile data supports role-aware navigation and database policies.

Field	Type	Key / rule	Description
`id`	UUID	PK, FK to `auth.users.id`, cascade delete	Stable identifier for the authenticated profile.
`email`	TEXT	Required, unique	Email associated with the profile and authentication account.
`username`	TEXT	Nullable, unique	Optional username shown or used by supported account workflows.
`full_name`	TEXT	Required; defaults to empty text	Account holder's display name.
`role`	TEXT	Required; `admin` or `member`; defaults to `member`	Application role used with authorization rules.
`account_status`	TEXT	Present in the live catalog where deployed	Account availability state when this optional live-schema field is enabled.
`created_at`	TIMESTAMPTZ	Required; defaults to current time	Time at which the profile was created.
`updated_at`	TIMESTAMPTZ	Present in the live catalog where deployed	Time of the most recent profile update when maintained by the deployed schema.

`members` — Approved Member and Current Membership

This table is the current operational record for an approved member. It contains identity and contact data, the current plan and dates, and member declarations collected in the application. `auth_user_id` associates the row with a sign-in identity when one exists; a partial unique index enforces no more than one member row for a non-null Auth ID. `application_id` can preserve the source registration link where the deployed schema has that relationship. Attendance, sales, and membership history refer to the member row rather than copying the entire profile.

Field	Type	Key / rule	Description
`id`	UUID	PK; generated when omitted	Identifier referenced by operational records.
`auth_user_id`	UUID	Nullable, unique FK to `auth.users.id`; deletion sets null	Optional link to the member's login identity.
`application_id`	UUID	Nullable FK to `membership_applications.id` in the deployed relationship	Registration application associated with creation of the member record.
`full_name`	TEXT	Required	Member's recorded full name.
`email`	TEXT	Nullable	Contact email copied from the application or account.
`username`	TEXT	Nullable	Member-facing account name where available.
`mobile`	TEXT	Nullable	Member's mobile contact number.
`date_of_birth`	DATE	Nullable	Date of birth supplied by the applicant.
`age`	SMALLINT	Nullable; nonnegative when supplied	Age recorded at application or registration time.
`sex`	TEXT	Nullable	Sex response supplied through the form.

Field	Type	Key / rule	Description
`height`	NUMERIC(7,2)	Nullable	Numeric height measurement.
`height_unit`	TEXT	Nullable; `cm`, `in`, or `ft`	Unit associated with the height value.
`weight_kg`	NUMERIC(7,2)	Nullable	Body weight in kilograms when provided.
`emergency_contact`	JSONB	Required; defaults to an empty object	Structured emergency-contact name, relationship, and contact details.
`plan`	TEXT	Nullable	Current membership plan label.
`plan_price`	NUMERIC(12,2)	Required; nonnegative; defaults to zero	Current plan charge recorded for the member.
`start_date`	DATE	Nullable	First date of the current membership period.
`end_date`	DATE	Present in compatible live schemas	End date retained by deployments using the `end_date` alias.
`expiry_date`	DATE	Nullable	Current membership expiry date used by the application.
`status`	TEXT	Required; constrained values in the schema	Membership state such as active, expired, or inactive.

Field	Type	Key / rule	Description
registration_source	TEXT	Required; defaults to Online application	Channel recorded for the member's registration.
paid	BOOLEAN or deployed payment value	Required in base schema; live projects may retain a compatible numeric representation	Indicates or records the payment state/value according to the deployed schema version.
fitness_goals	TEXT[]	Required; defaults to an empty array	One or more fitness goals supplied by the member.
health_conditions	TEXT[]	Required; defaults to an empty array	Selected health-condition declarations.
other_medical_conditions	TEXT	Required; defaults to empty text	Additional health information entered by the applicant.
signature_path	TEXT	Nullable	Stored path or data reference for the captured signature.
created_at	TIMESTAMPTZ	Required; defaults to current time	Member-record creation time.
updated_at	TIMESTAMPTZ	Required in the base schema; maintained on update	Most recent member-record update time.

membership_applications — Registration and Renewal Requests

Application rows preserve a submitted request and its review state. The application is distinct from the current member record: submission alone does not activate a membership. A registration request can be linked to the member created after approval, while a renewal request can identify an existing member. Review timestamps and reviewer identity make the decision traceable. This table includes health and emergency-contact data and should therefore be protected by the configured access policies.

Field	Type	Key / rule	Description
`id`	UUID	PK; generated when omitted	Identifier for the submitted request.
`auth_user_id`	UUID	FK to `auth.users.id`; required in the base schema	Auth identity that submitted the request.
`member_id`	UUID	Nullable FK to `members.id`; deletion sets null	Existing or resulting member associated with the application.
`application_type`	TEXT	Required; `registration` or `renewal`	Distinguishes a new membership request from a renewal request.
`full_name`	TEXT	Required	Applicant's or renewing member's name as submitted.
`email`	TEXT	Required in the base schema	Email contact supplied with the request.
`username`	TEXT	Nullable	Requested or associated username.
`mobile`	TEXT	Nullable	Mobile number supplied with the request.
`date_of_birth`	DATE	Nullable	Applicant's date of birth.
`age`	SMALLINT	Nullable; nonnegative when supplied	Age at the time of submission.
`sex`	TEXT	Nullable	Sex response supplied in the form.
`height`	NUMERIC(7,2)	Nullable	Height measurement supplied by the applicant.
`height_unit`	TEXT	Nullable; `cm`, `in`, or `ft`	Unit of the submitted height.
`weight_kg`	NUMERIC(7,2)	Nullable	Weight in kilograms if supplied.
`emergency_contact`	JSONB	Required; defaults to an empty object	Structured emergency contact details.
`emergency_contact_data`	JSONB	Required in compatible schema; defaults to an empty object	Compatibility field used by deployed form and migration versions.

Field	Type	Key / rule	Description
plan	TEXT	Required	Plan requested by the applicant.
plan_price	NUMERIC(12,2)	Required; nonnegative	Proposed plan charge recorded with the request.
start_date	DATE	Nullable	Requested or planned membership start date.
end_date	DATE	Nullable	Requested membership end date.
fitness_goals	TEXT[]	Required; defaults to an empty array	Goals selected in the application.
health_conditions	TEXT[]	Required; defaults to an empty array	Health declarations selected in the application.
other_medical_conditions	TEXT	Required; defaults to empty text	Additional health declaration text.
signature_path	TEXT	Nullable	Reference to the submitted signature.
status	TEXT	Required; pending, approved, or rejected; defaults to pending	Current review result.
submitted_at	TIMESTAMPTZ	Required; defaults to current time	Submission timestamp.
reviewed_at	TIMESTAMPTZ	Nullable	Time when an administrator reviewed the request.
reviewed_by	UUID	Nullable FK to auth.users.id; deletion sets null	Auth identity of the reviewing administrator.

attendance — Staff-Recorded Visits

An attendance row represents one visit period. The member relationship is required and uses a restrictive delete rule so that a member with attendance history cannot be silently removed while those records remain. time_out stays empty while a visit is open. The database check prevents a checkout timestamp earlier than check-in. The interface formats timestamps in Manila local time, while the database stores timestamp values.

Field	Type	Key / rule	Description
id	UUID	PK; generated when omitted	Identifier for the visit record.

Field	Type	Key / rule	Description
member_id	UUID	Required FK to members.id; deletion restricted	Member associated with the visit.
date	DATE	Present in live catalog where maintained	Calendar date used by date-based attendance views.
time_in	TIMESTAMPTZ	Required; defaults to current time	Timestamp at which staff records check-in.
time_out	TIMESTAMPTZ	Nullable; cannot precede time_in	Timestamp at which staff records checkout; null means visit remains open.
recorded_by	UUID	Nullable FK to auth.users.id; deletion sets null	Staff or administrator identity that recorded the event.
created_at	TIMESTAMPTZ	Required; defaults to current time	Time the attendance row was inserted.

products — Catalog and Inventory

Product rows describe items offered by the gym. The category and unit must agree: supplements are priced and sold by gram, while merchandise is sold by item and has tracked stock. Archiving marks a product inactive and records an archive timestamp in the live schema. Existing sale lines keep their own item description and pricing if a product is later archived or its catalog details change.

Field	Type	Key / rule	Description
id	UUID	PK; generated when omitted	Product identifier referenced by sale lines.
name	TEXT	Required	Product display name.
category	TEXT	Required; supplement or merchandise	Product class used by sales and inventory workflows.
price_per_unit	NUMERIC(12,2)	Required; nonnegative	Price per gram or per item according to unit.
unit	TEXT	Required; gram or item	Unit used to calculate a product sale.
size	TEXT	Nullable	Merchandise size or variant label.
stock	INTEGER	Nullable; nonnegative when present	Merchandise quantity available; supplement stock is not tracked in the base rule.
image_path	TEXT	Nullable	Product image reference shown in the catalog.
is_active	BOOLEAN	Required; defaults to true	Controls whether the product is available in active views.

Field	Type	Key / rule	Description
archived_at	TIMESTAMPTZ	Present in the live catalog; nullable	Time at which the product was archived.
created_at	TIMESTAMPTZ	Required; defaults to current time	Product creation timestamp.
updated_at	TIMESTAMPTZ	Required; defaults to current time	Most recent product update timestamp.

sales — Transaction Headers

The sales table stores one receipt-level transaction. It identifies the customer and processor, captures the total and status, and records when the sale occurred. A sale may refer to a member or represent a walk-in purchase. Detailed goods and services belong in sale_items, allowing one receipt to contain multiple lines. Receipt numbers are unique so they can be used to retrieve the corresponding transaction.

Field	Type	Key / rule	Description
id	UUID	PK; generated when omitted	Internal transaction identifier.
receipt_number	TEXT	Required, unique	Human-readable receipt reference.
customer_type	TEXT	Required; member or walk_in	Customer classification for the transaction.
customer_name	TEXT	Required	Customer name printed on the receipt.
member_id	UUID	Nullable FK to members.id; deletion sets null	Member link when the customer is a member.
total	NUMERIC(12,2)	Required; nonnegative	Receipt-level transaction total.
status	TEXT	Required; paid or voided; defaults to paid	Payment/void state used by reports.
sold_at	TIMESTAMPTZ	Required; defaults to current time	Transaction timestamp.
processed_by	UUID	Nullable FK to auth.users.id; deletion sets null	Auth identity of the administrator or staff account processing the sale.

sale_items — Itemized Receipt Lines

Each row is one service or product line on a sale. The sale relationship is required and cascades when a sale is removed. product_id is optional so membership lines and retained historical item descriptions do not require an active catalog product. The item name, category, unit, quantity, and price preserve what was sold at transaction time.

Field	Type	Key / rule	Description
`id`	UUID	PK; generated when omitted	Identifier for the receipt line.
`sale_id`	UUID	Required FK to `sales.id`; cascade delete	Receipt header containing this line.
`product_id`	UUID	Nullable FK to `products.id`; deletion sets null	Catalog product when the line represents a product sale.
`item_name`	TEXT	Required	Description shown on the receipt, retained independently of later catalog edits.
`category`	TEXT	Required; `supplement`, `merchandise`, or `membership`	Type of line item.
`quantity`	NUMERIC(12,3)	Required; greater than zero	Grams, item count, or membership quantity sold.
`unit`	TEXT	Required; `gram`, `item`, or `membership`	Quantity unit for the line.
`unit_price`	NUMERIC(12,2)	Required; nonnegative	Charge for each recorded unit.
`line_total`	NUMERIC(12,2)	Required; nonnegative	Total charged for this receipt line.

`membership_history` — Membership Lifecycle Events

History rows preserve changes to a member's subscription over time. Depending on the event, previous and new values may be present or null. This makes it possible to distinguish expiration snapshots from renewals and plan changes while retaining dates and prices. The member link is required and restrictive so historical rows are not discarded by deleting a member record.

Field	Type	Key / rule	Description
`id`	UUID	PK; generated when omitted	Identifier for the lifecycle event.
`member_id`	UUID	Required FK to `members.id`; deletion restricted	Member whose membership changed.
`event_type`	TEXT	Required; `expired`, `renewal`, or `plan_change`	Type of lifecycle event recorded.
`previous_plan`	TEXT	Nullable	Plan before the event.
`new_plan`	TEXT	Nullable	Plan after the event when applicable.
`previous_price`	NUMERIC(12,2)	Nullable; nonnegative	Prior plan price.

Field	Type	Key / rule	Description
`new_price`	NUMERIC(12,2)	Nullable; nonnegative	New plan price when applicable.
`previous_start_date`	DATE	Nullable	Start date of the prior membership period.
`previous_expiry_date`	DATE	Nullable	Expiry date of the prior membership period.
`new_start_date`	DATE	Nullable	Start date of the new period when applicable.
`new_expiry_date`	DATE	Nullable	Expiry date of the new period when applicable.
`notes`	TEXT	Required; defaults to empty text	Additional context associated with the event.
`recorded_at`	TIMESTAMPTZ	Required; defaults to current time	Time the history event was stored.
`recorded_by`	UUID	Nullable FK to `auth.users.id`; deletion sets null	Auth identity that recorded the event.

Referential Integrity and Record Lifecycle

The keys in the dictionary describe how the database preserves related records. A profile is dependent on its Auth identity. Applications can exist before their member relationship is established, then retain review information after a decision. Attendance and membership-history rows remain tied to their member. Sale items belong to a receipt and preserve their sold description even if the source product later disappears. These rules let reports combine current records with historical transactions without treating a submitted application as an approved membership.

The main lifecycle begins when an Auth identity submits a registration request. The request is stored as pending; an authorized review either records rejection or creates/links the member record and transaction through the configured workflow. A renewal request points to the existing member and stays pending until review. When staff record a renewal or plan change, the system preserves prior and new membership values in history and updates the current member row. Attendance events and sale headers are independent operational records that link back to the member when appropriate. Product rows are deactivated on archive so previous receipt lines can continue to describe the sale.

Foreign keys and `NOT NULL`, `CHECK`, and unique constraints provide database-level safeguards for identifiers, allowed values, quantities, and references. Row Level Security and privileged database functions control who may perform operations; the table definitions alone do not

grant access. Actual behavior also depends on the policies and functions deployed in the Supabase project.

FORMS

Form or screen	User	Purpose and principal inputs
Sign-in (`index.html`)	Applicant, member, administrator	Collects credentials for Supabase Auth and role-based routing.
Registration (`register.html`)	Applicant	Collects account and membership application details for administrator review.
Application review	Administrator	Displays application details and records approval or rejection.
Member portal (`customer.html`)	Member	Displays membership information, QR code, attendance history, active catalog, renewal request, and supported profile settings.
Attendance scanner	Administrator or staff using an administrator account	Scans member QR code and records check-in/check-out.
Product and inventory forms	Administrator	Creates or updates products, prices, units, active status, and merchandise stock.
Walk-in POS and receipt	Administrator	Records walk-in customer and sale-item details; displays receipt information.
Report filters	Administrator	Selects report type and available filters/date range; views or prints/saves results.
Password recovery (`reset-password.html`)	Account holder	Supports the password-reset workflow.

Form and Screen Descriptions

The screens below follow the user's task from entry through its result. Required fields and controls vary by workflow. The descriptions explain the purpose of each interface and how it connects to the records described in the Data Dictionary.

Sign-In Screen

The sign-in page accepts the account email and password and passes authentication to Supabase Auth. After successful authentication, the application reads the associated profile role and opens the appropriate member or administrator interface. When a valid account has a pending application rather than an approved member record, the application status is surfaced to the user. Invalid credentials, unavailable connectivity, or a missing profile should produce a recoverable message rather than imply that an application was approved. The page does not collect a role choice that could grant administrator access.

Registration Screen

The registration screen collects account credentials and membership application information. Depending on the form configuration, this includes name, date of birth, contact and emergency-contact details, selected plan, goals, health declarations, and digital signature. The form validates required values and formats before sending the request. Successful submission creates a pending application for administrator review; it does not directly grant membership access. The request is associated with the Auth identity so the applicant can later sign in and see the request state.

Application Review Screen

The administrator approval queue separates new registrations from renewal requests. Each row presents the submitted information needed for review and offers approve or reject actions. Approval of a registration creates or links the member record and records its membership sale where the transaction workflow succeeds. Approval of a renewal updates the existing membership period and records the renewal transaction/history. A rejected request receives a rejected status and reviewer/time information. These actions require an authenticated administrator and database-side authorization; hiding a button is not an access-control measure.

Administrator Dashboard and Membership Management

The dashboard presents operational summaries such as member count, daily attendance, sales, low merchandise stock, and pending applications. The values are derived from the records loaded for the current view and reflect the data available to the signed-in administrator. Membership management supports filtering current members, reviewing member details, changing a plan, and recording a counter renewal. A plan change or renewal updates the current membership fields while preserving the prior period in membership history or archive records. The resulting receipt or confirmation is displayed after the required database operations succeed.

Attendance Scanner

The attendance interface requests camera permission and scans a QR code issued for a member. The application resolves the identifier to a member record, checks whether an open visit already exists, and records check-in or check-out accordingly. An expired membership is not checked in, while an open visit can still be checked out. The scanner reports an unrecognized code, camera denial, network failure, or database error to staff. A member can view permitted attendance history in the portal, but the staff scanner is the workflow that creates and closes attendance events.

Product and Inventory Forms

The product form collects a name, category, unit price, and optional photo. Merchandise also collects size and stock quantity; supplements are priced by gram and do not use merchandise stock tracking in the base data model. The edit form updates the product catalog and can replace or remove the image. Archive changes the product to inactive and records the archive time, removing it from active catalog and sale selections while preserving historical sale-line descriptions. The interface validates prices and quantities before writing the row.

Point of Sale and Receipt

The point-of-sale form collects the customer name, selected product, and quantity. Merchandise quantity must be a whole number within available stock; supplement quantity may be fractional grams. A transaction is stored as a sale header with one or more itemized lines and an associated receipt number. The receipt shows customer, date, item description, unit, quantity, unit price, line total, and amount paid. Membership transactions use the membership category and do not require a product catalog link. The system should report a failed transaction clearly so that a displayed receipt is not mistaken for a saved sale.

Member Portal

The portal displays the signed-in member's current membership details and QR identifier, attendance history, and active product catalog where those features are enabled. Profile settings are limited to supported fields and apply only to the authenticated member's own profile. The renewal control is available only under the configured eligibility rule and creates an application for staff review. The portal should distinguish current membership information from a pending request so the user can see whether a renewal still needs approval.

Reports and Password Recovery

The reports screen lets the administrator choose a report and view or print its data; monthly export controls can package reports for a selected month. Password recovery uses the Supabase Auth recovery workflow and reset page. The reset screen accepts a replacement password only after a valid recovery session is established. It does not change membership or application status.

PROTOTYPE

The prototype consists of the interactive website pages and workflows described above. It includes login, registration, the member portal, administrator dashboard, attendance scanning, inventory, POS, receipts, and reports. The project files provide the implemented

pages and interfaces; screenshots can be inserted here if required by the school template. Do not claim stakeholder testing or user feedback unless those activities were conducted and recorded.

REPORTS

IronGrind provides administrative reports that present information already recorded by the system. The report view and PDF output use the same underlying operational records, with the selected report determining which records and fields are shown. A report can only be as complete as the stored transactions and member data. The reports are descriptive summaries and do not, by themselves, demonstrate an improvement in gym performance.

Membership Report

The membership report lists member name, plan, start date, and expiry date for current membership records in the active membership view. Members whose expired period has passed the configured archive grace period are excluded from that current list. The report helps staff review the current membership register; it is not a payment ledger and does not replace membership history.

Active and Inactive Member Report

The member-status report groups member records according to membership expiry and explicit inactive states such as expired, suspended, or disabled where recorded. It presents the member name, username, registration source, plan, and expiry. The group is a system-derived status view; staff should correct the underlying member record when a status is inaccurate.

Archives Report

The archives report combines expired membership snapshots, plan changes, and archived products. Membership entries preserve the previous plan period and its fee; plan-change entries show old and new plan information; product entries show the archived item and its catalog details. These categories have different meanings and are shown in separate subsections in the interactive view. An archive supports historical reference and does not issue refunds.

Payment and Sales Reports

The payment and sales reports list receipt number, customer, transaction total, transaction date, and status. Sale-item details are available through the receipt view and the itemized transaction table. Voided transactions should remain distinguishable from paid

transactions. Reconciliation should use the transaction records and the gym's approved financial process; the report is not an external accounting or payment-gateway statement.

Attendance Report

The attendance report groups visits by month and displays the member, date, check-in time, and check-out time. A missing checkout is shown as an open visit. Timestamp display is localized for the gym's Manila timezone. Attendance data records the events captured by staff and should not be interpreted as proof of access-control enforcement by a gate or turnstile.

Membership Renewal Report

The renewal report gathers completed renewal transactions from the counter and member-portal approval channels. It displays the receipt state, member, renewed plan, channel, amount, and date. If an approved legacy portal renewal has no corresponding sale row, the report can mark the missing receipt so staff can investigate rather than silently count it as recorded income. The renewal report depends on correct links between the application, member, and sale records.

Inventory Report

The inventory report is a snapshot of active products, category, unit price, and stock. Supplements are labeled by gram and merchandise by item; supplement stock is shown as not tracked when that is how the catalog is configured. The snapshot changes as products are edited, sold, or archived and does not represent a dated inventory audit unless one is separately saved.

Monthly Report Export

The monthly export prepares separate PDF documents for the available report types and packages them for download. Transactional reports filter dated records to the chosen month. Member-status and inventory reports represent their configured current snapshot; they are not historical snapshots unless the database stores dated state for that report. Archives use their event/archive dates. The export should be checked against the intended reporting period before it is submitted or used for reconciliation.

SYSTEM VERIFICATION AND EVALUATION

System verification and user evaluation serve different purposes. Verification checks whether each implemented workflow behaves according to its requirement. User evaluation examines the system's perceived quality and usefulness. No test results or user-evaluation

results are asserted in this chapter; actual outcomes must be collected and reported after the procedures are carried out.

Functional Verification

The following scenarios provide a minimum verification basis. The research team should record the test date, environment, test account role, observed result, and any defect for each executed scenario.

Test area	Verification scenario	Expected result
Authentication and access	Sign in with authorized administrator and member accounts and attempt role-restricted actions.	Each account reaches its intended interface; unauthorized database operations are denied by the configured policies.
Application workflow	Submit a registration application, review it as staff, and test both approval and rejection paths.	The application is recorded with its review status; approval creates or links the member record and associated transaction as configured.
Renewal	Submit a renewal request within the allowed period and review it as staff.	An eligible request is pending until staff decision; the membership is not renewed automatically.
Attendance	Scan a member QR code as staff for check-in and check-out, then view the member's history.	The event is recorded and the member can view their own history; the member cannot create the event through self-scan.
Inventory and sales	Record a merchandise sale and a gram-based supplement sale, then inspect the receipt and inventory records.	The sale and line items are saved with their units and totals; merchandise stock is updated according to the configured sales workflow.
Reports	Open and print or save available reports using records from the test data.	The report presents the corresponding stored records and totals for the selected report type.
Personal-data access	Attempt to read or update records using member and administrator accounts.	Members can access only their permitted personal data; administrator access follows the deployed policy configuration.

User Evaluation Framework

Chapter 2 identifies the DeLone and McLean Information Systems Success Model as a possible framework for evaluating system quality, information quality, use, user satisfaction, and net benefits. The evaluation should measure only dimensions included in the approved study instrument. The table below connects those dimensions to IronGrind and does not represent results.

Dimension	Possible IronGrind evaluation focus
System quality	Ease of navigating the browser workflows, reliability of common operations, and access behavior across member and administrator roles.

Dimension	Possible IronGrind evaluation focus
Information quality	Completeness, consistency, and usefulness of member, attendance, inventory, sales, and report information.
Service quality	Quality of support and assistance available to intended users during deployment and use, if this dimension is included in the approved instrument.
Use	Whether intended users can perform the workflows relevant to their role.
User satisfaction	Users' assessment of the system after completing representative tasks.
Net benefits	Perceived contribution to organizing the gym's records and operations; claims of improved efficiency or accuracy require evidence collected through the approved method.

Before administering an evaluation, the researchers should identify the actual participant population, sampling method, instrument, rating scale, consent procedure, and statistical treatment required by the approved proposal. These details and the resulting data are not present in the supplied project files and must not be invented. If the study uses a questionnaire, the instrument should be reviewed and approved before data collection, and the analysis should report the actual number of respondents and results.

SCOPE AND LIMITATIONS

IronGrind supports online membership applications and renewal requests for staff review; member access to selected membership and attendance information; staff-operated QR check-in and check-out; product and inventory records; in-person product and membership sales; receipts; and administrative operational reports. Its data model and workflows are tailored to the stated operations of Iron Grind Fitness Gym.

The project does not provide online payment processing, online product checkout or reservations, member-operated attendance entry, automated gate access, AI fitness coaching, or a native mobile application. The member QR code identifies the member for staff scanning; it does not itself grant gym access. The system's reports reflect the records entered and do not independently establish business outcomes. Data protection depends on correct deployment, configured policies, appropriate user practices, and the gym's operational procedures.

GANTT CHART

The development schedule follows the order of the Agile activities. The table records each phase, its main work, and its expected output. Calendar start and end dates should follow the approved project schedule.

Phase	Main activities	Expected output
1. Requirements planning	Define scope, users, workflows, data needs, and access requirements.	Prioritized requirements and acceptance criteria.
2. User design and prototyping	Refine page flows, forms, reports, database relationships, and role permissions.	Reviewed interface prototype and system diagrams.
3. Construction iterations	Implement prioritized features in the browser client and Supabase services; verify each increment.	Integrated feature increments and recorded issues.
4. Refinement	Resolve verification findings and update the prioritized requirements.	Refined system ready for deployment preparation.
5. Implementation and transition	Configure the intended environment, prepare administrator access, and orient staff.	Configured deployment and user-orientation records.
6. Evaluation	Conduct the approved user evaluation and analyze collected responses.	Evaluation dataset and reported findings.

Exact calendar dates and durations are not included because no approved dated schedule was present in the project files. Add those dates from the project plan before submitting a date-based Gantt chart.